# Olympiad Test Paper — Mathematics (Class 7) — Paper 3

## Chapter 4: Expressions Using Letter-Numbers

**Paper 3 — (progressively harder)**

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| **Subject** | Mathematics (NCERT Class 7) |
| **Chapter** | 4 — Expressions Using Letter-Numbers |
| **Total Questions** | 60 (Multiple Choice, single correct) |
| **Maximum Marks** | 240 |
| **Suggested Time** | 90 minutes |

**Marking scheme: +4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

**Instructions:** Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: reading 2n as n+2 (doubling vs adding two), evaluating in the wrong order (e.g. squaring a sum vs summing squares, applying −sign before the square), merging UNLIKE terms ($3a+4b$ becoming $7ab$, or $2a+3a^2$ becoming $5a^3$), forgetting a constant when combining like terms, mishandling the sign of a negative substitution, distributing a coefficient incompletely over a bracket, and confusing coefficient with constant. Keep every sign exactly as given; only LIKE terms combine; this chapter builds and simplifies expressions — it never solves equations. Do not write on this paper — mark answers on your answer sheet.

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**1.** A plumber charges a fixed visit fee of 80 rupees plus 45 rupees for each hour of work. If h is the number of hours worked, the total charge in rupees is:

(A) $80h + 45$
(B) $125h$
(C) $45h + 80$
(D) $45h - 80$

**2.** Reena is twice as old as her brother now. In 4 years she will be (using b for her brother's present age):

(A) $2(b + 4)$
(B) $b + 4$
(C) $2b + 4$
(D) $2b - 4$

**3.** Evaluate $3a^2 - 2a + 5$ when $a = 4$.

(A) 141
(B) 45
(C) 21
(D) 37

**4.** If $m = -3$, the value of $5 - m^2$ is:

(A) 14 (B) −14
(C) −4 (D) 4

**5.** Simplify $7x - 3y + 2 - 4x + 5y - 6$.

(A) $3x + 2y + 8$ (B) $3x + 2y - 4$
(C) $11x + 2y - 4$ (D) $3x - 2y - 4$

**6.** In the expression $9 - 4t$, which statement is correct?

(A) The coefficient of t is −4 and the constant term is 9
(B) The coefficient of t is 4 and the constant term is 9
(C) The coefficient of t is 9 and the constant term is −4
(D) The coefficient of t is −4 and the constant term is −9

**7.** Which pair below are LIKE terms?

(A) $5x^2y$ and $5xy^2$ (B) $-6xy$ and $11xy$
(C) $3ab$ and $3a$ (D) $7m$ and $7m^2$

**8.** Simplify $2(3x + 4) + 3(x - 2)$.

(A) $9x + 14$ (B) $9x - 2$
(C) $5x + 2$ (D) $9x + 2$

**9.** A taxi rule charges 30 rupees for the first kilometre and 12 rupees for each additional kilometre. For a trip of d kilometres ($d \geq 1$), the fare in rupees is:

(A) $12(d - 1) + 30$ (B) $12d + 30$
(C) $30(d - 1) + 12$ (D) $42d$

**10.** Evaluate $2(p + q)^2$ when $p = 1$ and $q = 3$.

(A) 20 (B) 16
(C) 32 (D) 10

**11.** If $a = 2$ and $b = -5$, the value of $a - b^2$ is:

(A) 27 (B) 23
(C) −27 (D) −23

**12.** A square pattern uses dots: figure 1 has 4 dots, figure 2 has 8, figure 3 has 12, increasing by 4 each time. A correct rule for the nth figure and its 15th value are:

(A) $n + 4$, giving 19 (B) $4n + 4$, giving 64
(C) $4n$, giving 60 (D) $4n$, giving 64

**13.** Simplify $6a - [2a - (3a - a)]$.

(A) 2a (B) 6a
(C) 8a (D) 10a

**14.** A rectangle has length $3x$ and breadth $(x + 2)$. Its perimeter, simplified, is:

(A) $4x + 2$ (B) $8x + 4$
(C) $8x + 2$ (D) $6x^2 + 4$

**15.** A growing border uses $2n + 6$ bricks for the nth ring. How many MORE bricks does ring 8 need than ring 5?

(A) 2 (B) 8
(C) 6 (D) 22

**16.** There are some marbles in a bag. If x is the number of marbles, the statement 'after adding 3 more, the total is doubled' describes which expression for the final count compared with the start? The final count is:

(A) $2x$ (B) $2x + 3$
(C) $x + 3$ (D) $2(x + 3)$

**17.** Simplify $5m + 3n - (2m - 4n)$.

(A) $3m - n$ (B) $3m + 7n$
(C) $7m - n$ (D) $3m + 7n^2$

**18.** The cost of n notebooks at 25 rupees each, with a flat 40-rupee discount on the whole order, is:

(A) $25(n - 40)$ (B) $25n - 40$
(C) $40 - 25n$ (D) $25n + 40$

**19.** In the expression $8 + 5x - x^2$, which is the constant term?

(A) 8 (B) 5
(C) $-1$ (D) $5x$

**20.** Simplify $3(2a - b) - 2(a - 2b)$.

(A) $4a - 5b$ (B) $4a - 7b$
(C) $4a + b$ (D) $8a + b$

**21.** Simplify $4x + 2x^2 - x + 5x^2 + 3x$.

(A) $13x^2 + 6x$ (B) $7x^3 + 6x$
(C) $13x^4$ (D) $7x^2 + 6x$

**22.** If $x = 3$ and $y = -2$, evaluate $4x - 3y + xy$.

(A) 12 (B) 24
(C) 6 (D) 18

**23.** Simplify $4(x + 3) - (x + 3)$.

(A) $3x + 3$ (B) $3x + 9$
(C) $4x + 9$ (D) $3x + 12$

**24.** A school orders n footballs at 350 rupees each and $(n + 5)$ cones at 20 rupees each. The total cost in rupees, simplified, is:

(A) $370n + 20$ (B) $370n^2 + 100$
(C) $350n + 25$ (D) $370n + 100$

**25.** If $k = -2$, the value of $3 - 2k^2$ is:

(A) 11 (B) −5
(C) 5 (D) 19

**26.** Simplify $7a - 5 - 3a + 8 - 2a$.

(A) $2a - 3$ (B) $2a + 3$
(C) $12a + 3$ (D) $2a + 13$

**27.** Matchstick triangles share sides: figure 1 uses 3 sticks, figure 2 uses 5, figure 3 uses 7, adding 2 each time. The rule for the nth figure is:

(A) $2n + 3$ (B) $3n$
(C) $2n + 1$ (D) $n + 2$

**28.** Evaluate $(2a - b)^2$ when $a = 3$ and $b = 1$.

(A) 35 (B) 25
(C) 5 (D) 37

**29.** Which expression is EQUIVALENT to $6x + 9$?

(A) $3(2x + 3)$ (B) $3(2x + 9)$
(C) $6(x + 9)$ (D) $3(2x) + 9x$

**30.** A bus has s seats per row and r rows, plus 4 extra standing places. The total capacity is:

(A) $rs + 4$ (B) $r + s + 4$
(C) $4rs$ (D) $rs - 4$

**31.** A pattern: stage 1 has 1 square, stage 2 has 4, stage 3 has 9, stage 4 has 16. The rule and the number of squares at stage 6 are:

(A) $2n$, giving 12 (B) $4n$, giving 24
(C) $n^2$, giving 12 (D) $n^2$, giving 36

**32.** If $a = 4$, the values of $2a^2$, $(2a)^2$ and $2a + 2$ are, respectively:

(A) 64, 64, 10 (B) 32, 64, 10
(C) 32, 32, 16 (D) 16, 64, 10

**33.** Simplify $10 - 3(2 - x)$.

(A) $3x + 16$ (B) $-3x + 4$
(C) $3x - 4$ (D) $3x + 4$

**34.** The cost of hiring a hall is 500 rupees plus 60 rupees per guest. Written as a rule, total cost C for g guests is:

(A) C = 500g + 60 (B) C = 60g + 500
(C) C = 560g (D) C = 60 + 500g

**35.** Simplify 2(a + b) + 3(a − b) − (a + b).

(A) 4a + 2b (B) 6a − 2b
(C) 4a − 4b (D) 4a − 2b

**36.** If p = 5 and q = 2, the value of $p^2 - 2pq + q^2$ is:

(A) 49 (B) 9
(C) 29 (D) −11

**37.** Three friends share a bill: the first pays x rupees, the second pays 20 more than the first, the third pays twice the first. The total they pay, simplified, is:

(A) 4x + 20 (B) 3x + 20
(C) 4x + 40 (D) 2x + 20

**38.** Which statement about 3a and $a^3$ is TRUE?

(A) 3a means a + a + a, while $a^3$ means a × a × a — they are unlike
(B) Both equal a × a × a (C) Both equal a + a + a
(D) They are like terms and combine to 4a

**39.** Evaluate $100 - n^3$ when n = 4.

(A) 88 (B) 64
(C) 36 (D) −36

**40.** A sequence has nth term 3n − 2. Which term of the sequence equals 25? (Find n that gives 25 by checking.)

(A) 9th (B) 8th
(C) 27th (D) 23rd

**41.** Simplify 5xy − 2x + 3xy + 4x − xy.

(A) 7xy + 2x (B) 9xy + 2x
(C) 7xy + 6x (D) $9x^2y^2$

**42.** If a = 6 and b = 3, the value of (a + b)/3 + ab is:

(A) 27 (B) 9
(C) 20 (D) 21

**43.** Twice a number is increased by 7, and the result is tripled. Using x for the number, this is:

(A) 2x + 7 (B) 3(2x) + 7
(C) 2(3x + 7) (D) 3(2x + 7)

**44.** If the perimeter of a square is given by 4s, then writing the perimeter of a square whose side is (s + 3) gives, simplified:

(A) $4s + 3$ (B) $4s + 12$
(C) $s + 12$ (D) $4s + 7$

**45.** Simplify $(5a - 2a) + (3a \times 2) - a$.

(A) $6a$ (B) $10a$
(C) $4a$ (D) $8a$

**46.** A worker earns w rupees on an ordinary day and $(w + 150)$ rupees on a holiday. In a month with 22 ordinary days and 3 holidays, total earnings, simplified, are:

(A) $25w + 150$ (B) $22w + 450$
(C) $25w + 3$ (D) $25w + 450$

**47.** Which expression is NOT equivalent to $12x - 8$?

(A) $4(3x - 2)$ (B) $2(6x - 4)$
(C) $4(3x - 4)$ (D) $12x - 8$

**48.** A rule gives output = $2 \times (\text{input})^2 - 5$. When the input is 3, the output is:

(A) 13 (B) 31
(C) 1 (D) 67

**49.** Simplify $6 + 2(3x - 1) - 4x$.

(A) $2x + 8$ (B) $2x + 5$
(C) $10x + 4$ (D) $2x + 4$

**50.** A stack of boxes: the bottom layer has 5 boxes and each layer above has 2 fewer than the layer below it. Using L for the number of layers is awkward; instead, the number of boxes in the layer that is 3 above the bottom is:

(A) $5 - 6 = -1$, so no such layer exists (the stack cannot have it)
(B) $5 + 6 = 11$ (C) $5 - 3 = 2$
(D) $5 \times 3 = 15$

**51.** If $x = -1$, the value of $x^3 + x^2 + x + 1$ is:

(A) 4 (B) 2
(C) 0 (D) −2

**52.** A garden path of length n metres is paved with 2 slabs per metre, and there are 5 extra slabs at the entrance. If each slab costs 30 rupees, the total cost in rupees is:

(A) $30(2n + 5)$ (B) $30(2n) + 5$
(C) $60n + 5$ (D) $30 \times 2n \times 5$

**53.** Simplify $8y - \{3y - (2y - y)\}$.

(A) $4y$ (B) $8y$
(C) $6y$ (D) $2y$

**54.** The number of legs on c chairs (4 legs each) and t three-legged tables, together, is:

(A) $4c + 3t$ (B) $7ct$
(C) $4c + 3t = 7(c + t)$ (D) $12ct$

**55.** If $a = 2$, $b = 3$, $c = 4$, evaluate $a + b \times c - a^2$.

(A) 16 (B) 20
(C) 10 (D) 6

**56.** Simplify $3p + 2q - 4 + 5 - p - 2q$.

(A) $2p + 1$ (B) $2p - 1$
(C) $2p + 4q + 1$ (D) $4p + 1$

**57.** Four expressions are evaluated at $x = 5$: $2x + 3$, $x^2 - 8$, $4x - 6$, $3x$. Which has the GREATEST value?

(A) $4x - 6$ (value 14) (B) $3x$ (value 15)
(C) $2x + 3$ (value 13) (D) $x^2 - 8$ (value 17)

**58.** A car rental costs a fixed 1200 rupees plus 9 rupees per kilometre. For a journey of k kilometres, which expression gives the cost in rupees, and what is it for $k = 100$?

(A) $1200k + 9$, giving 120009 (B) $9k + 1200$, giving 1308
(C) $9k + 1200$, giving 2100 (D) $1209k$, giving 120900

**59.** Simplify $a^2 + 2a - 3a^2 + a + 4$.

(A) $-2a^2 + 3a$ (B) $4a^2 + 3a + 4$
(C) $-2a^3 + 3a + 4$ (D) $-2a^2 + 3a + 4$

**60.** A pattern of L-shaped tiles: figure 1 needs 3 tiles, figure 2 needs 5, figure 3 needs 7. How many tiles in total are used in figures 1, 2 and 3 together, and what is the nth-figure rule?

(A) 15 tiles total; rule $3n$ (B) 12 tiles total; rule $2n + 1$
(C) 21 tiles total; rule $2n + 1$ (D) 15 tiles total; rule $2n + 1$

# Solutions — Mathematics (Class 7), Chapter 4: Expressions Using Letter-Numbers — Paper 3

**Paper 3 — (progressively harder)**

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

## Answer Key

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | C | 11 | D | 21 | D | 31 | D | 41 | A | 51 | C |
| 2 | C | 12 | C | 22 | A | 32 | B | 42 | D | 52 | A |
| 3 | B | 13 | B | 23 | B | 33 | D | 43 | D | 53 | C |
| 4 | C | 14 | B | 24 | D | 34 | B | 44 | B | 54 | A |
| 5 | B | 15 | C | 25 | B | 35 | D | 45 | D | 55 | C |
| 6 | A | 16 | C | 26 | B | 36 | B | 46 | D | 56 | A |
| 7 | B | 17 | B | 27 | C | 37 | A | 47 | C | 57 | D |
| 8 | D | 18 | B | 28 | B | 38 | A | 48 | A | 58 | C |
| 9 | A | 19 | A | 29 | A | 39 | C | 49 | D | 59 | D |
| 10 | C | 20 | C | 30 | A | 40 | A | 50 | A | 60 | D |

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

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## Worked Solutions

**1. (C)** The per-hour cost is 45 for each of h hours, giving 45h, and the fixed fee 80 is added once. Total = 45h + 80. '80h + 45' wrongly multiplies the fixed fee by h and adds the per-hour rate once. '125h' wrongly merges 45 and 80 (they are unlike — one is per-hour, one is fixed). '45h − 80' subtracts the visit fee instead of adding it.

**2. (C)** Reena's present age is 2 × b = 2b. In 4 years add 4: 2b + 4. '2(b + 4)' = 2b + 8 doubles the brother's future age, not Reena's present-plus-4. 'b + 4' uses the brother's age, not twice it. '2b − 4' subtracts instead of adds the 4 years.

**3. (B)** Square first: $a^2 = 16$, so $3a^2 = 48$. Then 2a = 8. So 48 − 8 + 5 = 45. '141' comes from doing $(3a)^2 = (12)^2 = 144$ then −8+5, i.e. squaring 3a instead of a. '21' comes from treating $3a^2$ as 3a = 12, giving 12 − 8 + 5 (wrong, ignores the square) — actually 12−8+5=9, but 21 arises from (3·4 + 2·4 + 5) sign error; either way the square was dropped. '37' subtracts 2a after the +5 is mis-ordered.

**4. (C)** First $m^2 = (-3)^2 = 9$ (a square is never negative). Then $5 - 9 = -4$. '14' comes from treating $-m^2$ as $+9$ wrongly as $5 + 9$, or reading $5 - m$ as $5-(-3)=8$ then... no: $14 = 5 - (-9)$, i.e. wrongly making $m^2$ negative. '−14' takes $(5 - m)^2$ ordering errors. '4' comes from $m^2 = (-3)^2$ mis-squared to 1 giving $5-1$.

**5. (B)** Like terms: x-terms $7x - 4x = 3x$; y-terms $-3y + 5y = 2y$; constants $2 - 6 = -4$. Result $3x + 2y - 4$. '$3x + 2y + 8$' mis-adds the constants as $2 - (-6)$. '$11x + 2y - 4$' adds $7x + 4x$ instead of subtracting. '$3x - 2y - 4$' mishandles $-3y + 5y$ as $-2y$.

**6. (A)** The term $-4t$ carries the minus sign with its coefficient, so the coefficient of t is $-4$; the term 9 with no letter is the constant. The first distractor drops the sign of the coefficient. The second swaps coefficient and constant. The third wrongly attaches a minus to the constant 9.

**7. (B)** Like terms have exactly the same letters with the same powers; $-6xy$ and $11xy$ both have $xy$. '$5x^2y$ and $5xy^2$' differ in which letter is squared. '3ab and 3a' differ because one lacks b. '$7m$ and $7m^2$' differ in the power of m.

**8. (D)** Distribute: $2(3x+4) = 6x + 8$ and $3(x-2) = 3x - 6$. Add: $(6x+3x) + (8-6) = 9x + 2$. '$9x + 14$' adds $8 + 6$ instead of $8 - 6$. '$9x - 2$' takes $8 - 6$ as $-2$... it computes constants as $-8+6$ wrongly. '$5x + 2$' forgets to multiply $3x$ by nothing — it adds $6x$ and... actually drops one term, giving $6x - x$ type slip.

**9. (A)** The first km costs 30 (fixed); the remaining $(d - 1)$ km cost 12 each, giving $12(d - 1)$; total $12(d - 1) + 30$. '$12d + 30$' charges 12 for ALL d km including the first, then adds 30, double-counting the first km. '$30(d-1)+12$' swaps the two rates. '$42d$' wrongly merges 30 and 12 and multiplies by d.

**10. (C)** Inside the bracket first: $p + q = 4$. Square: $4^2 = 16$. Multiply by 2: 32. '20' comes from $2(p^2 + q^2) = 2(1 + 9) = 20$, squaring each term separately instead of the sum. '16' forgets the factor 2. '10' is $p^2 + q^2 = 1 + 9$ without squaring the sum or multiplying by 2.

**11. (D)** First $b^2 = (-5)^2 = 25$. Then $a - b^2 = 2 - 25 = -23$. '27' comes from $2 - (-25)$, i.e. wrongly making $b^2$ negative. '23' is $25 - 2$, subtracting in the wrong order. '−27' comes from $-2 - 25$ with a sign slip on a.

**12. (C)** Differences are constant at 4 and figure 1 = 4, so the rule is $4n$. For $n = 15$: $4 \times 15 = 60$. '$n + 4$' gives $1+4=5$ for $n=1$, not 4. '$4n + 4$' gives 8 for $n=1$, not 4. '$4n$ giving 64' uses the right rule but $4 \times 15 = 60$, not 64.

**13. (B)** Innermost: $3a - a = 2a$. Next: $2a - (2a) = 0$. Then $6a - 0 = 6a$. '2a' stops after $6a - 4a$, mishandling the nested signs. '8a' adds the bracket instead of subtracting. '10a' adds everything ignoring brackets.

**14. (B)** Perimeter = 2(length + breadth) = $2(3x + x + 2) = 2(4x + 2) = 8x + 4$. '$4x + 2$' is just length + breadth, not doubled. '$8x + 2$' doubles $4x$ but forgets to double the +2. '$6x^2 + 4$' wrongly multiplies $3x$ by $(x+2)$.

**15. (C)** Ring 8: $2(8)+6 = 22$. Ring 5: $2(5)+6 = 16$. Difference $22 - 16 = 6$. (Quick check: each ring adds 2 bricks, over 3 rings that is $3 \times 2 = 6$.) '2' is the per-ring increase, not over three rings. '8' adds 6 + 2 confusing the formula. '22' is the count for ring 8 itself, not the difference.

**16. (C)** Adding 3 more to $x$ simply gives $x + 3$ marbles in the bag. The phrase 'the total is doubled' is a misleading description — adding 3 does not double anything unless $x = 3$; the actual final count is $x + 3$. '$2x$' would mean the count was literally doubled. '$2x + 3$' and '$2(x + 3)$' both invent a doubling that adding 3 does not perform.

**17. (B)** Remove the bracket, changing both inner signs: $5m + 3n - 2m + 4n$. Combine: $(5m-2m) + (3n+4n) = 3m + 7n$. '$3m - n$' keeps the inner signs unchanged $(3n - 4n)$. '$7m - n$' adds $5m + 2m$. '$3m + 7n^2$' wrongly raises n to a power when adding like terms.

**18. (B)** n notebooks cost $25n$; subtract the single 40-rupee discount: $25n - 40$. '$25(n - 40)$' subtracts 40 from the count of books. '$40 - 25n$' reverses the subtraction. '$25n + 40$' adds the discount instead of subtracting it.

**19. (A)** The constant term is the one with no letter, namely 8. '5' is the coefficient of x. '−1' is the coefficient of $x^2$. '$5x$' is a term containing the variable, not a constant.

**20. (C)** Distribute: $3(2a-b) = 6a - 3b$; $-2(a-2b) = -2a + 4b$. Combine: $(6a - 2a) + (-3b + 4b) = 4a + b$. '$4a - 5b$' fails to flip the sign of $-2b$ (gives $-3b - 4b$... ). '$4a - 7b$' adds $-3b - 4b$. '$8a + b$' adds $6a + 2a$.

**21. (D)** $x^2$-terms: $2x^2 + 5x^2 = 7x^2$. x-terms: $4x - x + 3x = 6x$. Result $7x^2 + 6x$ ($x^2$ and x are UNLIKE, so they stay separate). '$13x^2 + 6x$' wrongly throws all coefficients onto $x^2$. '$7x^3 + 6x$' adds the powers when combining like terms. '$13x^4$' merges everything into one impossible term.

**22. (A)** $4x = 12$; $-3y = -3(-2) = +6$; $xy = (3)(-2) = -6$. Sum: $12 + 6 - 6 = 12$. '24' comes from $-3y$ read as $-6$ wrongly added as... or xy taken as +6. '6' takes xy as +6 and $-3y$ as $-6$. '18' drops the xy term.

**23. (B)** There are 4 lots of $(x+3)$ minus 1 lot, leaving 3 lots: $3(x+3) = 3x + 9$. Or expand: $(4x+12) - (x+3) = 3x + 9$. '$3x + 3$' subtracts the constants wrongly $(12 - 9)$. '$4x + 9$' subtracts only the constant, not the x. '$3x + 12$' forgets to subtract the +3.

**24. (D)** Footballs: $350n$. Cones: $20(n + 5) = 20n + 100$. Total: $350n + 20n + 100 = 370n + 100$. '$370n + 20$' forgets to multiply 5 by 20. '$370n^2 + 100$' wrongly squares n. '$350n + 25$' fails to distribute 20 over $(n+5)$ and merges counts.

**25. (B)** $k^2 = (-2)^2 = 4$; $2k^2 = 8$; $3 - 8 = -5$. '11' comes from $2k^2$ read as $2(-2) = -4$ then $3 - (-4)$ wrongly squared, or making $k^2$ negative: $3 - (-8)$. '5' computes $k^2$ as 1. '19' takes $(3 - 2k)^2$ ordering errors.

**26. (B)** a-terms: $7a - 3a - 2a = 2a$. Constants: $-5 + 8 = 3$. Result $2a + 3$. '$2a - 3$' mis-signs the constants. '$12a + 3$' adds all a-coefficients as if positive. '$2a + 13$' adds 5 + 8 ignoring the minus on 5.

**27. (C)** Constant difference 2 gives 2n plus a fixed part; for $n = 1$, $2(1) + 1 = 3$ ✓; $n = 2$ gives 5 ✓. The rule is $2n + 1$. '$2n + 3$' gives 5 for n=1. '$3n$' gives 6 for n=2. '$n + 2$' gives 3 for n=1 but 4 for n=2, not 5.

**28. (B)** Inside first: $2a - b = 6 - 1 = 5$. Square: $5^2 = 25$. '35' comes from $4a^2 - b^2 = 36 - 1$, squaring terms separately and dropping the cross part. '5' forgets to square. '37' is $4a^2 + b^2 = 36 + 1$, wrong expansion.

**29. (A)** $3(2x + 3) = 6x + 9$ ✓ (factor out 3). '$3(2x + 9)$' $= 6x + 27$. '$6(x + 9)$' $= 6x + 54$. '$3(2x) + 9x$' $= 6x + 9x = 15x$, which adds an extra x term.

**30. (A)** Seats fill r rows × s seats = rs, plus 4 standing: $rs + 4$. '$r + s + 4$' adds rows and seats instead of multiplying. '4rs' wrongly multiplies the standing places into the seat count. '$rs - 4$' subtracts the standing places.

**31. (D)** The counts 1, 4, 9, 16 are $1^2, 2^2, 3^2, 4^2$, so the rule is $n^2$; at $n = 6$ it is 36. '2n' gives 2,4,6,8 — wrong from stage 1. '4n' gives 4 at stage 1, not 1. '$n^2$ giving 12' uses the right rule but $6^2 = 36$, not 12.

**32. (B)** $2a^2 = 2 \times 16 = 32$ (square a first). $(2a)^2 = 8^2 = 64$ (the whole 2a is squared). $2a + 2 = 8 + 2 = 10$. '64,64,10' wrongly squares 2a in the first. '32,32,16' fails to square the bracket and mis-adds. '16,64,10' computes $2a^2$ as $a^2$ leaving out the 2.

**33. (D)** Distribute the −3: $10 - 6 + 3x = 4 + 3x = 3x + 4$. The −3 times −x gives +3x. '$3x + 16$' adds 6 instead of subtracting. '$-3x + 4$' fails to flip the sign on −x. '$3x - 4$' mis-signs the constant ($10 - 6 = 4$, not −4).

**34. (B)** Per-guest cost 60 applies to g guests (60g); the 500 is fixed and added once: $C = 60g + 500$. '$C = 500g + 60$' multiplies the fixed fee by g. '$C = 560g$' merges unlike quantities. '$C = 60 + 500g$' makes the per-guest rate fixed and the fixed fee per-guest.

**35. (D)** Expand: $2a+2b + 3a-3b - a-b = (2a+3a-a) + (2b-3b-b) = 4a + (-2b) = 4a - 2b$. '$4a + 2b$' mishandles the b-signs. '$6a - 2b$' adds the a's as 2+3+1. '$4a - 4b$' over-subtracts a b.

**36. (B)** $p^2 = 25$; $2pq = 2 \cdot 5 \cdot 2 = 20$; $q^2 = 4$. So $25 - 20 + 4 = 9$. (Check: equals $(p-q)^2 = 3^2 = 9$.) '49' is $(p+q)^2 = 7^2$, wrong sign on the middle term. '29' drops the $-2pq$ subtraction

(25 + 4). '−11' subtracts $q^2$ instead of adding.

**37. (A)** First x; second x + 20; third 2x. Sum: x + (x+20) + 2x = 4x + 20. '3x + 20' forgets the second 'x' inside x+20... actually counts only x + 20 + 2x. '4x + 40' doubles the 20. '2x + 20' drops the third person's share partially.

**38. (A)** 3a is repeated addition (a+a+a) and $a^3$ is repeated multiplication (a·a·a); they are unlike terms and cannot be combined. 'Both equal a×a×a' confuses the two. 'Both equal a+a+a' likewise. 'combine to 4a' treats unlike terms as like.

**39. (C)** $n^3 = 4 \times 4 \times 4 = 64$; 100 − 64 = 36. '88' takes $n^3$ as 3n = 12. '64' gives just $n^3$ and forgets the 100 −. '−36' reverses to 64 − 100.

**40. (A)** Test by substitution: 3(9) − 2 = 27 − 2 = 25 ✓. '8th' gives 3(8)−2 = 22. '27th' gives 79. '23rd' gives 67. Only n = 9 yields 25.

**41. (A)** xy-terms: 5xy + 3xy − xy = 7xy. x-terms: −2x + 4x = 2x. Result 7xy + 2x (xy and x are unlike). '9xy + 2x' adds 5+3+1. '7xy + 6x' adds 2+4 ignoring the minus. '$9x^2y^2$' merges all into one impossible term.

**42. (D)** (a + b)/3 = 9/3 = 3; ab = 6 × 3 = 18; sum 3 + 18 = 21. '27' uses a/3 + b/3 + ab wrongly or a + b/3 + ... '9' computes only (a+b)/3 path errors then +6. '20' takes (a+b)/3 as 2 (a/3 only).

**43. (D)** Twice the number is 2x; increased by 7 gives (2x + 7); tripling the result gives 3(2x + 7). '2x + 7' stops before tripling. '3(2x) + 7' triples only the 2x, not the +7. '2(3x + 7)' swaps the doubling and tripling order wrongly.

**44. (B)** New side is (s + 3); perimeter = 4(s + 3) = 4s + 12. '4s + 3' forgets to multiply 3 by 4. 's + 12' multiplies only the 3 by 4 and leaves s alone. '4s + 7' adds 4 + 3 instead of 4 × 3.

**45. (D)** 5a − 2a = 3a; 3a × 2 = 6a; then 3a + 6a − a = 8a. '6a' computes 3a × 2 only and forgets the rest. '10a' adds 3a + 6a + a (wrong sign on last). '4a' takes 3a × 2 as 3a + 2 type slip.

**46. (D)** Ordinary: 22w. Holiday: 3(w + 150) = 3w + 450. Total: 22w + 3w + 450 = 25w + 450. '25w + 150' forgets to multiply 150 by 3. '22w + 450' drops the 3 holiday w's. '25w + 3' ignores the 150 bonus entirely.

**47. (C)** 4(3x − 4) = 12x − 16, which is NOT equal to 12x − 8, so it is the odd one out. '4(3x − 2)' = 12x − 8 ✓ equivalent. '2(6x − 4)' = 12x − 8 ✓ equivalent. '12x − 8' is identical. The question asks for the non-equivalent one.

**48. (A)** Input squared first: $3^2 = 9$; times 2 = 18; minus 5 = 13. '31' squares the doubled input: $(2·3)^2 − 5 = 36 − 5$. '1' computes 2×3 − 5 forgetting the square. '67' takes $2 \times (3^2)$ wrongly as 2 × 36.

**49. (D)** Distribute: $6 + 6x - 2 - 4x = (6x - 4x) + (6 - 2) = 2x + 4$. '$2x + 8$' adds $6 + 2$ instead of $6 - 2$. '$2x + 5$' mis-subtracts the constant. '$10x + 4$' adds $6x + 4x$ instead of subtracting.

**50. (A)** Each layer up removes 2 boxes: 3 layers up removes $3 \times 2 = 6$, giving $5 - 6 = -1$, an impossible count, so that layer cannot exist. '$5 + 6 = 11$' adds instead of subtracts. '$5 - 3 = 2$' subtracts the number of layers instead of 2 per layer. '$5 \times 3$' multiplies wrongly.

**51. (C)** $x^3 = (-1)^3 = -1$; $x^2 = 1$; $x = -1$; constant 1. Sum: $-1 + 1 - 1 + 1 = 0$. '4' treats every term as +1. '2' makes $x^3 = +1$ wrongly. '−2' makes $x^2 = -1$ wrongly.

**52. (A)** Number of slabs = $2n + 5$; cost = $30 \times (2n + 5) = 30(2n + 5) = 60n + 150$. '$30(2n) + 5$' charges 30 per path slab but only 5 rupees... it adds 5 slabs uncosted. '$60n + 5$' costs the path but adds 5 rupees not 5 slabs. '$30 \times 2n \times 5$' multiplies the entrance count in wrongly.

**53. (C)** Innermost: $2y - y = y$. Next: $3y - y = 2y$. Then $8y - 2y = 6y$. '$4y$' computes $8y - 4y$ mishandling nesting. '$8y$' subtracts 0 wrongly. '$2y$' subtracts $6y$.

**54. (A)** Chairs contribute $4c$ legs and tables $3t$ legs; since c-chairs and t-tables are unlike, the total stays $4c + 3t$. '$7ct$' merges unlike terms and multiplies the counts. 'The $7(c+t)$' claim is false since $4c+3t \neq 7c+7t$. '$12ct$' multiplies 4 and 3 and both letters.

**55. (C)** Multiplication first: $b \times c = 12$; $a^2 = 4$. Then $a + 12 - 4 = 2 + 12 - 4 = 10$. '16' adds $a + b$ first: $(a+b)\times c = 20$ then $-4$. '20' is $(a+b)\times c$ with no $a^2$. '6' computes $a + b + c - a^2$ type slip.

**56. (A)** p-terms: $3p - p = 2p$. q-terms: $2q - 2q = 0$. Constants: $-4 + 5 = 1$. Result $2p + 1$. '$2p - 1$' mis-signs the constants. '$2p + 4q + 1$' adds the q-terms instead of cancelling. '$4p + 1$' adds $3p + p$.

**57. (D)** At $x = 5$: $2x+3 = 13$; $x^2-8 = 25-8 = 17$; $4x-6 = 20-6 = 14$; $3x = 15$. The greatest value is 17, from $x^2-8$. '$4x-6$' gives 14. '$3x$' gives 15. '$2x+3$' gives 13. None of these beats 17.

**58. (C)** Cost = $9k + 1200$; at $k = 100$: $900 + 1200 = 2100$. '$1200k + 9$' multiplies the fixed fee by k. '$9k + 1200$ giving 1308' wrongly computes $9\times12$ instead of $9\times100$. '$1209k$' merges the fixed and per-km amounts.

**59. (D)** $a^2$-terms: $a^2 - 3a^2 = -2a^2$. a-terms: $2a + a = 3a$. Constant: 4. Result $-2a^2 + 3a + 4$. '$-2a^2 + 3a$' drops the constant. '$4a^2 + 3a + 4$' adds the $a^2$-terms as $1 + 3$. '$-2a^3 + 3a + 4$' raises the power wrongly.

**60. (D)** Total = 3 + 5 + 7 = 15. Rule: difference 2 with figure 1 = 3 gives 2n + 1 ($2 \cdot 1+1=3$ ✓). 'rule 3n' gives 6 for n=2, wrong. '12 tiles' mis-adds 3+5+7. '21 tiles' adds an extra figure (3+5+7+... ) wrongly.